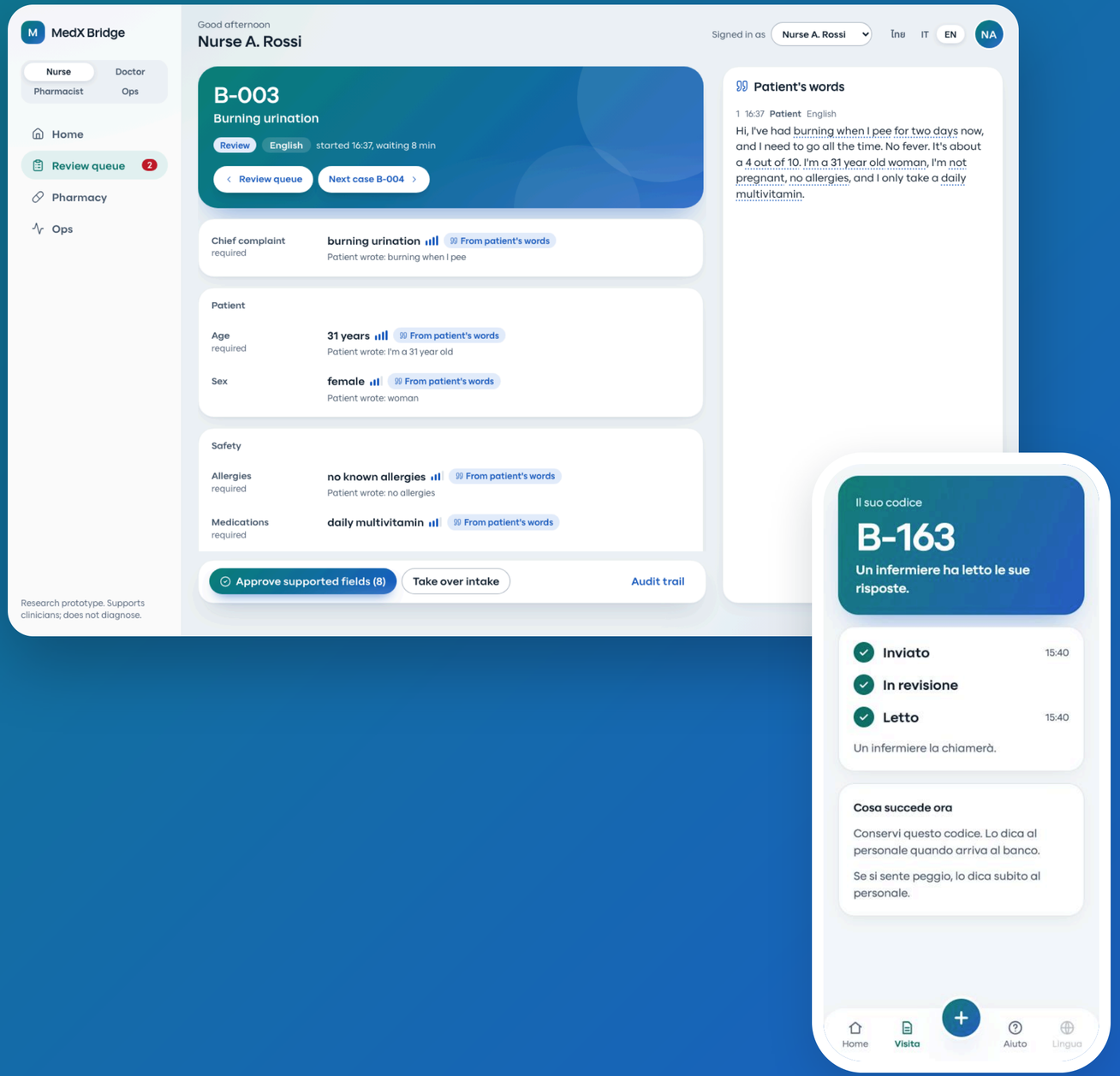
MedX Bridge

AMD Developer Hackathon ACT III · Track 2

MedX Bridge

Multilingual intake for nurses and pharmacists. Every field points back to the patient's own words.



Research prototype. Supports clinicians; does not diagnose.
Synthetic data only. github.com/KingPhuripol/medx-bridge

A language gap and a wrong medicine list are both safety events

49.1% vs 29.5%

Physical harm in adverse events, limited-English patients vs English speakers, six US hospitals. Communication errors caused 52.4% vs 35.9%.[1]

13–59%

Patients with at least one unintended error in their admission medication list across six studies; omission was the commonest type.[2]

49.2% vs 27.0%

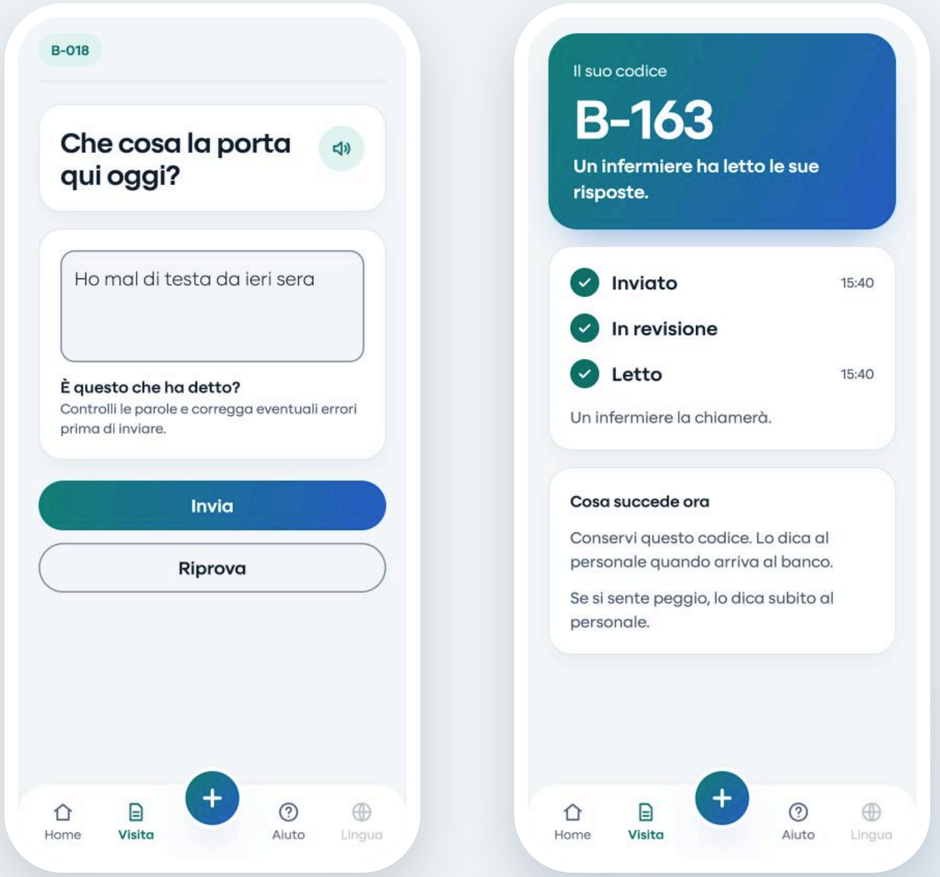
Doctor's day on EHR and desk work vs face time with the patient, four specialties.[3]

Thailand had 32.97 million foreign tourist arrivals in 2025.[4] At the desk, the patient and the staff often do not share a language.

1 Divi et al., Int J Qual Health Care 2007 (PMID 17277013). 2 Studies M1 to M6 in docs/research/PHARMACY-PROBLEM.md; settings differ, so a range is shown. 3 Sinsky et al., Ann Intern Med 2016 (PMID 27595430). 4 Ministry of Tourism and Sports via Bernama, 6 Jan 2026. Full list: docs/research/PROBLEM.md.

Three people, one intake, before the consult

Patient	Speaks or types in Thai, Italian or English. A voice agent reads the question aloud; the patient checks and corrects the transcript before it is sent.
Nurse	Reads an English summary (design goal: quick to scan). Clicks a field to see the patient's own sentence. Approves, edits or takes over.
Pharmacist	Reconciles the medicines the patient named, brand to generic, with alerts on duplicate paracetamol, allergy conflicts and interactions.
Clinic manager	Sees the queue by urgency, escalations, abstain rate and model latency.



Left: the patient confirms what the voice agent heard. Right: after the nurse approves, the patient's phone ticks "Reviewed" in the patient's language.

Real screenshots from the built app (Expo patient app on web, Italian UI). "2 to 3 minutes per intake" is a design assumption, not a cited figure.

Rules first, Gemma second, a human last

1 Redact
Names, phone, ID become tokens before any storage or model call.
code

→

2 Red-flag rules
18 rules, every turn. A hit stops the intake; no model can clear it.
code, no model

→

3a Gemma on AMD
Extract, translate, one quote per field.
model

3b Gemma safety screen
Add-only: raises, never clears.
model, in parallel

→

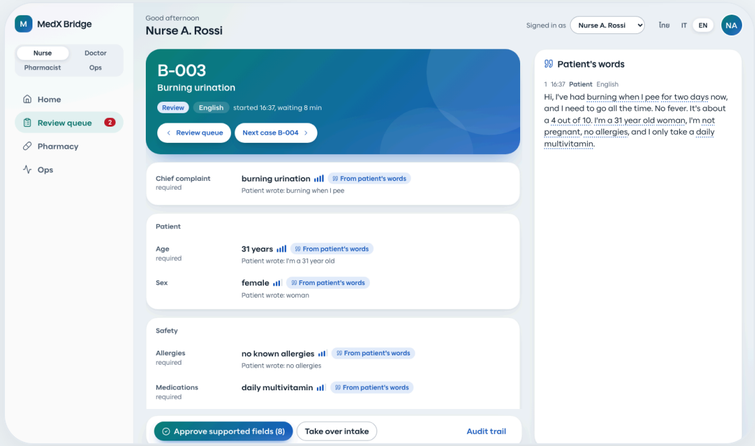
4 Verify
Code checks each quote exists. None: "needs check".
code

→

5 Ask or hand over
One follow-up question, or abstain.
model + code

→

6 Clinician
Approve, edit, take over. Audit log.
human



From patient's words

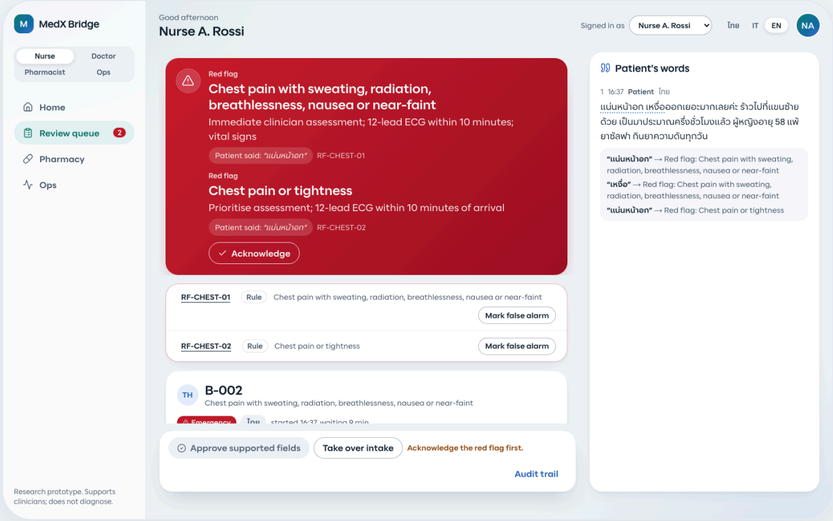
Each field in the nurse view carries the quote it came from, in the patient's own words, and a confidence bar. The highlighted sentence is the proof. Fields without a quote are shown as "needs check", never as fact.

Provider error, schema error or low confidence gives "needs human review", not a guess.

Design: docs/ai/DESIGN.md. Rules: docs/clinical/SPEC.md (18 rules, 175 reference tests). Screenshot: nurse review screen, synthetic case B-003.

Safety by design, in five answers

Wrong or unsupported answers	No verified quote means "needs check". Output is schema-checked; quotes must match the patient's words. Injection handling is not yet evaluated end to end.
Sensitive information	Redaction before storage or any model call. Synthetic data only. Keys stay server-side.
Uncertain results	Confidence is shown on every field. Low confidence or provider failure: abstain, "needs human review".
High-risk decisions	Rules run before any model. The Gemma screen is add-only, shown as "model-suggested, confirm".
Human review	Nothing is final until a clinician approves. Every action is in the append-only audit log.



Thai patient, chest pain and sweating in the first message: the nurse screen is interrupted, the patient's sentence is highlighted, the nurse acknowledges or marks a false alarm with a reason. The patient phone shows a calm "go to the desk now" screen.

Mechanisms and code paths: pitch/SAFETY.md. Cross-vendor review: Claude read-only reviewed the Codex engine (docs/reports/w6-engine-review.md) and QA scripts hit every route (docs/reports/w7-qa-bugs.md); findings fixed with regression tests; remaining limitations are listed in the submission.

Evidence we can show, and the gap that motivated the screen

What	Count	Where
Synthetic cases (40 intake + 12 medication)	**52**	data/cases
Blind holdout cases, written without seeing the rules	**18**	data/holdout
Red-flag rules / reference tests	**18 / 175**	docs/clinical
Drugs in dictionary / reference tests	**59 / 194**	docs/clinical
Automated tests, all passing	**1,093**	apps/web, vitest

Small synthetic blind holdout (18 cases), keyword rules alone

7 of 10

emergencies escalated; 3 of 8 non-emergencies raised a false alarm. Misses were lay wording that keyword lists cannot cover. That is why Gemma adds a safety screen.

With the Gemma screen

Holdout recall
[pending AMD run]

Asserted-field precision
[pending AMD run]

Unsupported rate, verifier on vs off
[pending AMD run]

Synthetic evaluation, research prototype, small n; not clinically validated. Holdout result: eval/results/holdout-rules-only.md, recorded before any change. Test count: npx vitest run in apps/web, 17 Oct 2026. Pre-declared metrics: docs/ai/METRICS.md.

Gemma via AMD Token Factory, behind one gateway

Clients

Patient app (Expo), clinician, pharmacist and ops web (Next.js)

→

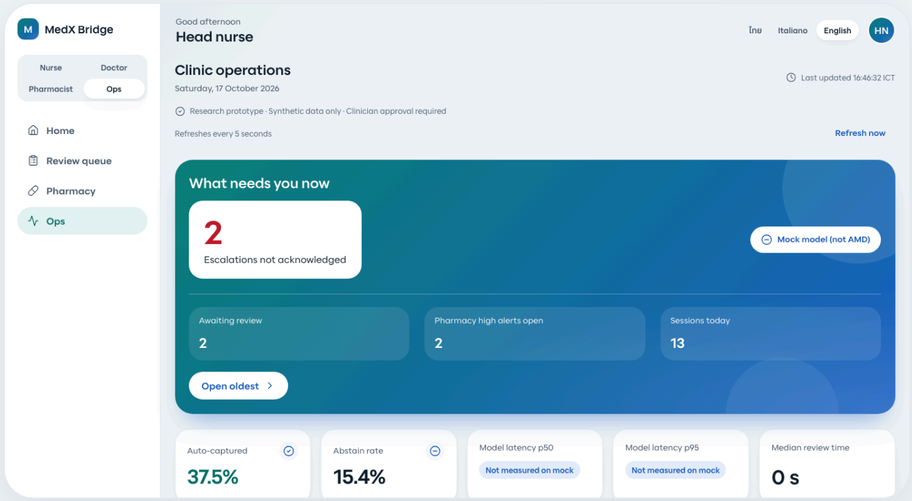
API and pipeline

Redact, rules, verify, audit. Provider SDKs stay inside one adapter.

→

AMD Token Factory

Designed to run Gemma on AMD. OpenAI-compatible; endpoint and model id come from environment variables.



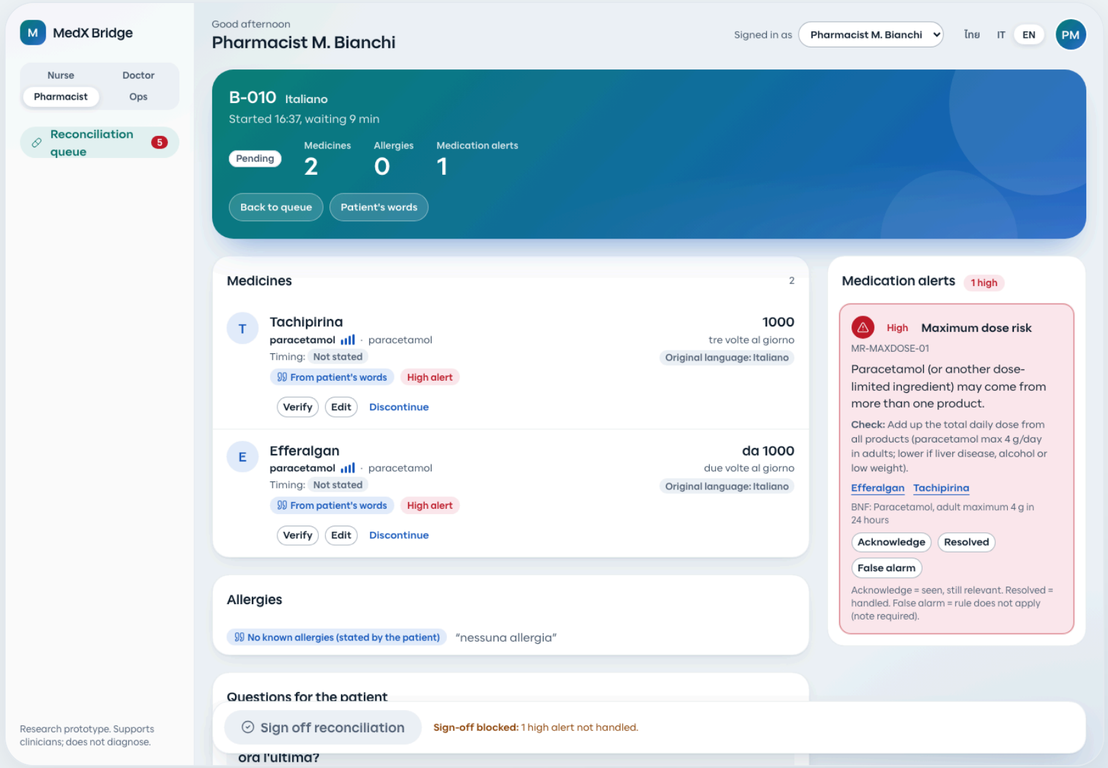
- Three jobs for Gemma (designed): extract and translate, add-only safety screen, next question.
- Current results use a mock model until the endpoint is connected; the ops screen says so ("Mock model, not AMD") instead of showing invented latency.
- Latency p50 / p95 per turn: **[pending AMD run]**
- Model variant chosen on a dev split: **[pending AMD run]**

Designed to run Gemma on AMD via Token Factory; the key was still pending, so current results use a mock model. Budget: turn p50 at most 6 s, p95 at most 12 s (docs/ai/DESIGN.md section 7), to be measured and reported either way.

Next: prove it with clinicians before anyone relies on it

- Run the pre-declared evaluation on Gemma on AMD and publish every result, including misses.
- Native-speaker review of Thai and Italian text.
- Timed study with nurses and pharmacists, then ethics approval before any real data.
- Not a medical device. Supports clinicians; never diagnoses, prescribes or decides urgency.

Open source, MIT.
github.com/KingPhuripol/medx-bridge
Team: [names and roles to add]



Screenshot: pharmacist reconciliation, synthetic case B-010: Tachipirina and Efferalgan both contain paracetamol, flagged as a maximum-dose risk; sign-off is blocked until the alert is handled by the pharmacist.